Brendan Reed brreed@andrew.cmu.edu

95-760 Decision Making Under Uncertainty Peter Zhang

Part A: Linear Program Formulations (Deterministic)

Decision Variables

- x1: Number of Aqua-Spas to produce.
- x2: Number of Hydro-Lux's to produce.
- y: Number of pumps to purchase.

Objective Function

Maximize total profit: 360(x1) + 300(x2) - 105(y)

Constraints

- **Labor:** 9(x1) + 6(x2) ≤ 1,572
- **Tubing:** 12(x1) + 16(x2) ≤ 2,940
- **Pump:** x1 + x2 ≤ y
- **Non-negativity:** x1, x2, y ≥ 0

Part B: Stochastic Programming Model (Under Uncertainty)

Scenario Probabilities

Scenario (s)	Hydro-Lux Labor (L2s)	Aqua-Spa Profit (P1s)	Tubing Available (Ts)	Joint Probability (ps)
1	5 hrs (prob. 0.5)	$320 (prob. 0.6)	2,400 ft (prob. 0.1)	0.5 x 0.6 x 0.1 = **0.03**
2	5 hrs (prob. 0.5)	$320 (prob. 0.6)	3,000 ft (prob. 0.9)	0.5 x 0.6 x 0.9 = **0.27**
3	5 hrs (prob. 0.5)	$420 (prob. 0.4)	2,400 ft (prob. 0.1)	0.5 x 0.4 x 0.1 = **0.02**
4	5 hrs (prob. 0.5)	$420 (prob. 0.4)	3,000 ft (prob. 0.9)	0.5 x 0.4 x 0.9 = **0.18**
5	7 hrs (prob. 0.5)	$320 (prob. 0.6)	2,400 ft (prob. 0.1)	0.5 x 0.6 x 0.1 = **0.03**
6	7 hrs (prob. 0.5)	$320 (prob. 0.6)	3,000 ft (prob. 0.9)	0.5 x 0.6 x 0.9 = **0.27**
7	7 hrs (prob. 0.5)	$420 (prob. 0.4)	2,400 ft (prob. 0.1)	0.5 x 0.4 x 0.1 = **0.02**
8	7 hrs (prob. 0.5)	$420 (prob. 0.4)	3,000 ft (prob. 0.9)	0.5 x 0.4 x 0.9 = **0.18**

Decision Variables

- **First-Stage:**
  - y: Number of pumps to purchase.
- **Second-Stage** (where s is from 1 to 8):
  - x1s: Number of Aqua-Spas to produce in scenario s.
  - x2s: Number of Hydro-Lux's to produce in scenario s.

Objective Function

Maximize the *expected* profit by subtracting the guaranteed cost of pumps from the probability-weighted sum of the profit across all 8 scenarios.

Maximize: $\mathbf{-105(y) + \Sigma_s\, p_s(P1_s \cdot x1_s + 300 \cdot x2_s)}$ **for s = (1,2,3,4,5,6,7,8)**

Constraints

- **Labor (for s = 1 to 8):**
  9(x1s) + L2s(x2s) ≤ 1,572
- **Tubing (for s = 1 to 8):**
  12(x1s) + 16(x2s) ≤ Ts
- **Pump (for s = 1 to 8):**
  x1s + x2s ≤ y
- **Non-negativity:**
  y ≥ 0, x1s ≥ 0, x2s ≥ 0 for all s

Part D: Optimal Solution Comparison

Part A Expected Profit: $47,177

Part C Expected Profit: $46,531

The deterministic model yields a higher expected profit of $47,177 because it assumes all uncertain parameters take their expected average values, allowing Howie to optimize production without controlling against variability. The stochastic model's lower expected profit of $46,531 reflects the true cost of uncertainty. Howie must commit to a pump quantity before knowing which scenario will realize, so the first-stage decision is more conservative to remain feasible across all eight possible outcomes.

95-760 Decision Making Under Uncertainty		**Brendan Reed**
Peter Zhang	**HW4**	**brreed@andrew.cmu.edu**

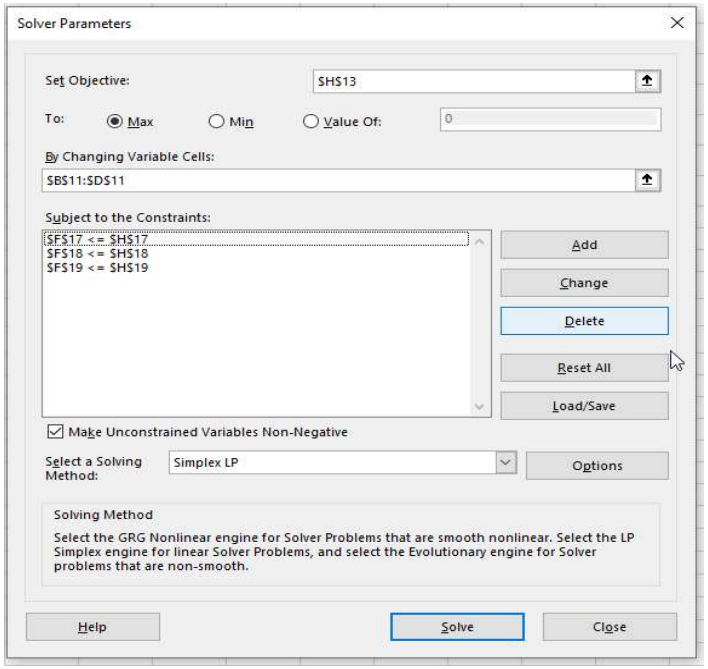
Part A: Deterministic Model

	Aqua-Spa (x1)	Hydro-Lux (x2)	Pumps (y)
Decision Variables	104.3333333	105.5	209.833333

Objective (Profit)	360	300	-105

Total Profit:	47177.5

Constraints					Used	Sign	Available
Labor (hrs)	9	6	0		1572	<=	1572
Tubing (ft)	12	16	0		2940	<=	2940
Pumps	1	1	-1		0	<=	0



	95-760 Decision Making Under Uncertainty	**Brendan Reed**
Peter Zhang	**HW4**	**brreed@andrew.cmu.edu**

Part A: Deterministic Model

	Aqua-Spa (x1)	Hydro-Lux (x2)	Pumps (y)
Decision Variables	104.333333333333	105.5	209.833333333333

Objective (Profit)	360	300	-105

Total Profit:	=SUMPRODUCT(B11:D11,B13:D13)

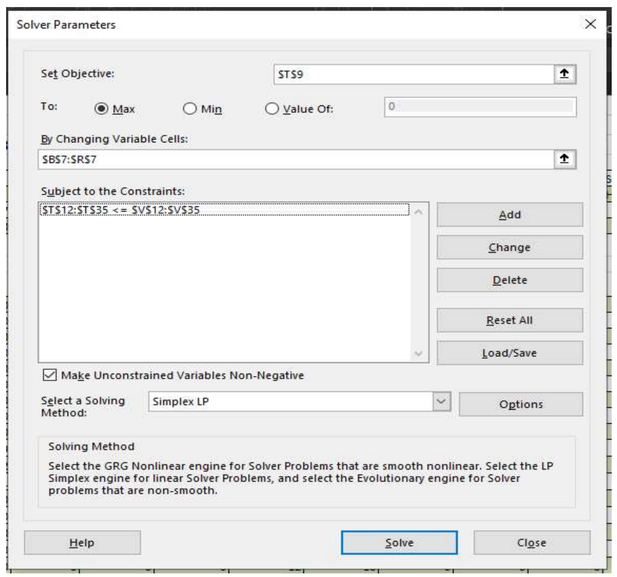
Constraints					Used	Sign	Available
Labor (hrs)	9	6	0		=SUMPRODUCT($B$11:$D$11,B17:D17)	<=	1572
Tubing (ft)	12	16	0		=SUMPRODUCT($B$11:$D$11,B18:D18)	<=	2940
Pumps	1	1	-1		=SUMPRODUCT($B$11:$D$11,B19:D19)	<=	0

Part C: Stochastic Program Model

Probabilities	**1**	**0.03**	**0.03**	**0.27**	**0.27**	**0.02**	**0.02**	**0.18**	**0.18**	**0.03**	**0.03**	**0.27**	**0.27**	**0.02**	**0.02**	**0.18**	**0.18**
	Stage 1	Scen 1	Scen 1	Scen 2	Scen 2	Scen 3	Scen 3	Scen 4	Scen 4	Scen 5	Scen 5	Scen 6	Scen 6	Scen 7	Scen 7	Scen 8	Scen 8
Variables	Pumps (y)	Aqua-Spa	Hydro-Lux	Aqua-Spa	Hydro-Lux	Aqua-Spa	Hydro-Lux	Aqua-Spa	Hydro-Lux	Aqua-Spa	Hydro-Lux	Aqua-Spa	Hydro-Lux	Aqua-Spa	Hydro-Lux	Aqua-Spa	Hydro-Lux
Decision Variables	204.8	156.5714286	32.57142857	137	67.8	156.5714286	32.57142857	137	67.8	139.2	45.6	69.2	135.6	174.6666667	0	174.6666667	0
Objective (Profit/Cost)	-105	320	300	320	300	420	300	420	300	320	300	320	300	420	300	420	300

Expected Profit: **46531.05714**

Constraints																		**Used**	**Sign**	**Available**
S1 Labor	0	9	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1572	<=	1572
S1 Tubing	0	12	16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2400	<=	2400
S1 Pumps	-1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-15.65714286	<=	0
S2 Labor	0	0	0	9	5	0	0	0	0	0	0	0	0	0	0	0	0	1572	<=	1572
S2 Tubing	0	0	0	12	16	0	0	0	0	0	0	0	0	0	0	0	0	2728.8	<=	3000
S2 Pumps	-1	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	<=	0
S3 Labor	0	0	0	0	0	9	5	0	0	0	0	0	0	0	0	0	0	1572	<=	1572
S3 Tubing	0	0	0	0	0	12	16	0	0	0	0	0	0	0	0	0	0	2400	<=	2400
S3 Pumps	-1	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	-15.65714286	<=	0
S4 Labor	0	0	0	0	0	0	0	9	5	0	0	0	0	0	0	0	0	1572	<=	1572
S4 Tubing	0	0	0	0	0	0	0	12	16	0	0	0	0	0	0	0	0	2728.8	<=	3000
S4 Pumps	-1	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	<=	0
S5 Labor	0	0	0	0	0	0	0	0	0	9	7	0	0	0	0	0	0	1572	<=	1572
S5 Tubing	0	0	0	0	0	0	0	0	0	12	16	0	0	0	0	0	0	2400	<=	2400
S5 Pumps	-1	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	-20	<=	0
S6 Labor	0	0	0	0	0	0	0	0	0	0	0	9	7	0	0	0	0	1572	<=	1572
S6 Tubing	0	0	0	0	0	0	0	0	0	0	0	12	16	0	0	0	0	3000	<=	3000
S6 Pumps	-1	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	-2.84217E-14	<=	0
S7 Labor	0	0	0	0	0	0	0	0	0	0	0	0	0	9	7	0	0	1572	<=	1572
S7 Tubing	0	0	0	0	0	0	0	0	0	0	0	0	0	12	16	0	0	2096	<=	2400
S7 Pumps	-1	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	-30.13333333	<=	0
S8 Labor	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	7	1572	<=	1572
S8 Tubing	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12	16	2096	<=	3000
S8 Pumps	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	-30.13333333	<=	0



Part C: Stochastic Program Model

Probabilities	**1**	**0.03**	**0.03**	**0.27**	**0.27**	**0.02**	**0.02**	**0.18**	**0.18**	**0.03**	**0.03**	**0.27**	**0.27**	**0.02**	**0.02**	**0.18**	**0.18**
	Stage 1	Scen 1	Scen 1	Scen 2	Scen 2	Scen 3	Scen 3	Scen 4	Scen 4	Scen 5	Scen 5	Scen 6	Scen 6	Scen 7	Scen 7	Scen 8	Scen 8
Variables	Pumps (y)	Aqua-Spa	Hydro-Lux	Aqua-Spa	Hydro-Lux	Aqua-Spa	Hydro-Lux	Aqua-Spa	Hydro-Lux	Aqua-Spa	Hydro-Lux	Aqua-Spa	Hydro-Lux	Aqua-Spa	Hydro-Lux	Aqua-Spa	Hydro-Lux
Decision Variables	204.8	156.571428571429	32.5714285714286	137	67.8	156.571428571429	32.5714285714286	137	67.8	139.2	45.6	69.2	135.6	174.666666666667	0	174.666666666667	0
Objective (Profit/Cost)	-105	320	300	320	300	420	300	420	300	320	300	320	300	420	300	420	300

Expected Profit:	**=SUMPRODUCT(B7:R7, B8:R8, B3:R3)**

Constraints																		**Used**	**Sign**	**Available**
S1 Labor	0	9	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	=SUMPRODUCT($B57:$R57,B12:R12)	<=	1572
S1 Tubing	0	12	16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	=SUMPRODUCT($B57:$R57,B13:R13)	<=	2400
S1 Pumps	-1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	=SUMPRODUCT($B57:$R57,B14:R14)	<=	0
S2 Labor	0	0	0	9	5	0	0	0	0	0	0	0	0	0	0	0	0	=SUMPRODUCT($B57:$R57,B15:R15)	<=	1572
S2 Tubing	0	0	0	12	16	0	0	0	0	0	0	0	0	0	0	0	0	=SUMPRODUCT($B57:$R57,B16:R16)	<=	3000
S2 Pumps	-1	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	=SUMPRODUCT($B57:$R57,B17:R17)	<=	0
S3 Labor	0	0	0	0	0	9	5	0	0	0	0	0	0	0	0	0	0	=SUMPRODUCT($B57:$R57,B18:R18)	<=	1572
S3 Tubing	0	0	0	0	0	12	16	0	0	0	0	0	0	0	0	0	0	=SUMPRODUCT($B57:$R57,B19:R19)	<=	2400
S3 Pumps	-1	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	=SUMPRODUCT($B57:$R57,B20:R20)	<=	0
S4 Labor	0	0	0	0	0	0	0	9	5	0	0	0	0	0	0	0	0	=SUMPRODUCT($B57:$R57,B21:R21)	<=	1572
S4 Tubing	0	0	0	0	0	0	0	12	16	0	0	0	0	0	0	0	0	=SUMPRODUCT($B57:$R57,B22:R22)	<=	3000
S4 Pumps	-1	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	=SUMPRODUCT($B57:$R57,B23:R23)	<=	0
S5 Labor	0	0	0	0	0	0	0	0	0	9	7	0	0	0	0	0	0	=SUMPRODUCT($B57:$R57,B24:R24)	<=	1572
S5 Tubing	0	0	0	0	0	0	0	0	0	12	16	0	0	0	0	0	0	=SUMPRODUCT($B57:$R57,B25:R25)	<=	2400
S5 Pumps	-1	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	=SUMPRODUCT($B57:$R57,B26:R26)	<=	0
S6 Labor	0	0	0	0	0	0	0	0	0	0	0	9	7	0	0	0	0	=SUMPRODUCT($B57:$R57,B27:R27)	<=	1572
S6 Tubing	0	0	0	0	0	0	0	0	0	0	0	12	16	0	0	0	0	=SUMPRODUCT($B57:$R57,B28:R28)	<=	3000
S6 Pumps	-1	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	=SUMPRODUCT($B57:$R57,B29:R29)	<=	0
S7 Labor	0	0	0	0	0	0	0	0	0	0	0	0	0	9	7	0	0	=SUMPRODUCT($B57:$R57,B30:R30)	<=	1572
S7 Tubing	0	0	0	0	0	0	0	0	0	0	0	0	0	12	16	0	0	=SUMPRODUCT($B57:$R57,B31:R31)	<=	2400
S7 Pumps	-1	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	=SUMPRODUCT($B57:$R57,B32:R32)	<=	0
S8 Labor	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	7	=SUMPRODUCT($B57:$R57,B33:R33)	<=	1572
S8 Tubing	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12	16	=SUMPRODUCT($B57:$R57,B34:R34)	<=	3000
S8 Pumps	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	=SUMPRODUCT($B57:$R57,B35:R35)	<=	0